HI^{2}D^{2}FNet: Hyperspectral Intrinsic Image Decomposition Guided Data Fusion Network for Hyperspectral and LiDAR Classification

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***Abstract*— In multimodal data fusion and land-cover interpretation tasks, the fusion interpretability between hyperspectral image (HSI) and light detection and ranging (LiDAR) data is always nontrivial to be clarified. Furthermore, the heterogeneous sample and distribution variances of these two remote sensing (RS) modalities impede the joint classification performance. In this article, a hyperspectral intrinsic image decomposition guided data fusion network (HI^{2}D^{2}FNet) is proposed. Generally, classic hyperspectral intrinsic image decomposition (HIID) performs well in image enhancement and shadow removal. It decomposes one HSI into one reflectance component and one shading component. Inspired by the core mechanism of HIID, our motivation is to preliminarily exploit its potential for multimodal RS data fusion and explore the inherent modality connection between HSI and LiDAR data from the intrinsic perspective. Specifically, compared with existing techniques, HI^{2}D^{2}FNet is capable of fusing the horizontal geometry information in the shading component with the vertical geometry information in the LiDAR data from both sample and distribution perspectives in the spatial domain. The generated cross-modal geometry feature contributes to guiding the reflectance stream optimization. This unique fusion framework connects both the modalities with respect to geometry information and enhances the specific fusion interpretability. The decomposition and fusion modules in HI^{2}D^{2}FNet are optimized simultaneously in a novel alternative optimization pattern. Furthermore, several unique cross-modal constraints in terms of prior RS properties are presented. Experiments conducted on three widely available datasets prove the superiority of HI^{2}D^{2}FNet over state-of-the-art techniques. The source codes will be available at https://github.com/GEOywb/HI2D2FNet.**

***Index Terms*— Classification, hyperspectral, intrinsic image decomposition, light detection and ranging (LiDAR), multimodality.**

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I. Introduction

REMOTE sensing (RS) techniques capture and analyze the physical characteristics of on-ground materials and objects from a distance [1], which reflect remarkable advantages in urban planning [2], precision agriculture [3], mapping annotation [4], etc. With the rapid development of aerospace satellites and spacecraft, an increasing amount of multimodal RS images, including hyperspectral images (HSIs) and light detection and ranging (LiDAR) data, can be obtained from diverse sensor platforms, describing objects within target regions at multiple levels [5]. Specifically, HSIs with hundreds of channels reflect detailed spatial and spectral information, covering the ultraviolet (UV), visible and near-infrared (VNIR), and shortwave infrared (SWIR) electromagnetic spectra [6], [7]. LiDAR data contain object elevations and describe geometry information along the vertical spatial direction [8], [9]. Both the RS modalities simulate diverse land-cover properties, providing the feasibility of precise interpretation [10]. Thus, jointly exploiting and fusing both the RS modalities has held tremendous potential in RS-related fields.

A substantial number of RS data fusion methods have been presented for hyperspectral and LiDAR joint classification tasks [11]. Classic data fusion techniques can be categorized into two groups, including feature-level fusion (FLF) and decision-level fusion (DLF) methods. Generally, FLF methods consist of one feature extraction module and one fusion module. The former with several feature extractors extracts high-level features from modalities. The latter fuses the obtained features for classification subsequently. Notably, the addition and concatenation operators are commonly chosen for fusion. Peng et al. [12] presented a multiview hierarchical network (MVHN) for high-quality fusion images. Zhang et al. [13] used the attention mechanism to enhance the feature extraction of HSI and LiDAR data. Kuras et al. [14] presented a novel approach to the multitemporal analysis of HSI and LiDAR data at the feature level. Four pathways were designed to extract multitemporal features from both the RS modalities. Zhang et al. [15] proposed a novel three-branch convolutional neural network (CNN) to extract spectral, spatial, and elevation features. Furthermore, they constructed a novel mutual guidance attention module for feature fusion. To enhance

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the fusion performance, Li et al. [16] presented a novel compatibility fusion strategy as a substitute for classic fusion strategies. Zhao et al. [17] replaced the classic concatenation layer with a novel cross-token attention fusion encoder. The aforementioned techniques prove that FLF methods have relatively high fusion flexibilities. Nevertheless, they always contain two separate streams for HSI and LiDAR data, making the feature extraction and fusion modules separated mutually. DLF methods design several classifiers and introduce unique voting strategies to generate ultimate classification results [18]. For instance, Ge et al. [19] proposed a Tikhonov-regularization-based kernel collaborative representation classification method. In view of the strong fusion robustness, they are always combined with FLF methods for better fusion performance [20], [21], [22].

Deep learning (DL) techniques have provided significant possibilities in hyperspectral and LiDAR fusion and joint classification (FJC) tasks recently [12], [23], [24]. In view of the fusion pattern, DL fusion contains three baselines [20], i.e., early fusion, middle fusion, and late fusion. Early fusion directly fuses modalities before feature extraction and outperforms other baselines when the modality variance is slight. Late fusion extracts high-level features initially for subsequent fusion and improves the modality distribution consistency. Middle fusion balances these two baselines and extracts joint features from fused high-level features. To analyze their advantages, Hong et al. [20] proposed a simple but effective fusion baseline called ENDnet, fusing HSI and LiDAR data in a more compact fashion. Furthermore, they provided a general multimodal DL network and multiple fusion strategies [25]. Hang et al. [26] considered dual coupled CNNs (CPCNNs) in their framework. Both the FLF and DLF methods were used simultaneously to fuse cross-modal features comprehensively. Li et al. [27] adopted two CNNs to extract high-level features from RS modalities and constructed an enhanced multisource fusion network called EMFnet for remarkable fusion performance.

Regarding the detailed fusion performance, these techniques achieve promising success practically. Nevertheless, constructing excellent data fusion networks for hyperspectral and LiDAR classification is hindered commonly due to two reasons: *fusion interpretability* and *heterogeneous variance*.

1) *Fusion Interpretability:* Existing data fusion methods tend to construct effective DL networks for high-level spectral, spatial, and elevation features from HSI and LiDAR data initially and fuse them afterward. Various cross-modal information transferring strategies are used for better fusion performance. However, they always treat HSI and LiDAR data as one common modality pair without clarifying their specific fusion interpretability. Furthermore, black-box DL networks further hamper the improvement of this interpretability. In this article, we preliminarily provide motivations and explanations for data fusion network construction from the intrinsic perspective considering the prior properties of these two RS modalities.
2) *Heterogeneous Variance:* Generally, multimodal FJC tasks require relatively high consistency between two RS modalities. However, the heterogeneous sample and distribution variances between HSI and LiDAR data remarkably impede the joint classification performance. Thus, capturing their inherent modality connection and simultaneously fusing both samples and distributions between HSI and LiDAR data become the primary tasks.

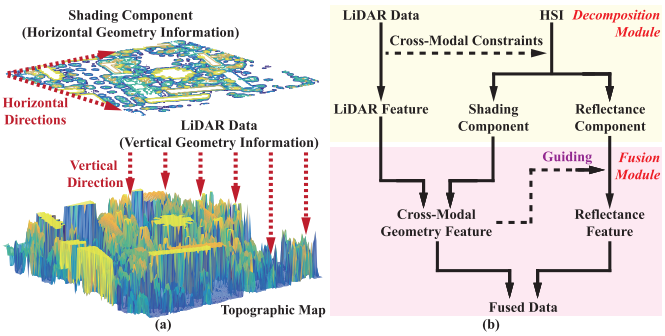

Fig. 1. Core novelty of HI^{2}D^{2}FNet for a clear and intuitive explanation. (a) Complementary property of the shading component and the LiDAR data with respect to geometry information. (b) Working principle from a high-level viewpoint. The feature extraction processes are not marked in (b).

To enhance the specific fusion interpretability, the core mechanism of hyperspectral intrinsic image decomposition (HIID) is considered for exploring the inherent modality connection between HSI and LiDAR data from the intrinsic perspective. Classic HIID has become an effective tool for image enhancement and shadow removal. However, its application in multimodal data fusion is weak and limited. Thus, our motivation is to preliminarily exploit its potential for HSI and LiDAR data fusion. Generally, HIID decomposes one single HSI into one reflectance component and one shading component [28]. The former reflects the object materials, which are invariant to illumination. The latter contains detailed illumination (e.g., shadowing, sky-lighting, street-lighting) in specific scenarios, which is related to the light wavelength and spatial arrangement. Kang et al. [29] introduced IID into RS-related fields for the first time to verify its potential. Jin and Gu [30] and Jin et al. [31] improved IID and further presented an effective HIID framework by removing the shading component directly. Moreover, they fused low-resolution HSIs with panchromatic images for image enhancement [32]. Apparently, the shading components were always treated as superfluous and removed directly for intraclass consistency and stability.

In our work, the practical application potential of the shading component is exploited. Specifically, this component contains abundant horizontal geometry information, describing the arrangement of objects along the horizontal directions. It is associated with not only the luminous intensity but also the geometry profile. Conversely, the LiDAR data record the object elevations along the vertical direction. Thus, they describe the geometry information of objects from two complementary perspectives and should be considered together [see Fig. 1(a)]. This article explores the potential relationships between the shading component and the LiDAR data from the intrinsic perspective and fuses them for cross-modal geometry features. Subsequently, these features with comprehensive geometry information are used for guiding the reflectance

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stream optimization. Compared with fusing high-level hyperspectral and LiDAR features by two separate streams, this work attempts to bridge the gap between HSI and LiDAR data from the intrinsic perspective by considering the core mechanism of HIID. The aforementioned decomposition and fusion procedures enhance the fusion interpretability. Besides, without using the existing HIID methods, multiple novel cross-modal constraints in terms of prior RS properties are considered in the whole network for better decomposition performance, further enhancing the decomposition and fusion interpretability.

Inspired by the core mechanism of HIID, we propose a hyperspectral intrinsic image decomposition guided data fusion network (HI^{2}D^{2}FNet), aiming to explore the inherent modality connection between HSI and LiDAR data from the intrinsic perspective to decrease the interference of their heterogeneous variance. Its working principle is shown in Fig. 1(b). HI^{2}D^{2}FNet extends the application of HIID and preliminarily exploits its potential for bridging the gap between these two RS modalities. Nevertheless, it does not directly use the existing HIID methods but considers the core mechanism of HIID in the whole fusion network. To further enhance the interpretability, HI^{2}D^{2}FNet presents multiple unique cross-modal constraints in terms of prior RS properties. The decomposition and fusion modules in HI^{2}D^{2}FNet are optimized in a novel alternative optimization pattern. This pattern attempts to achieve a dynamic balance between both the modules for better decomposed components and greater classification performance simultaneously. Specifically, the decomposition module is constrained to generate explainable and meaningful components, and the fusion module aims to tackle the heterogeneous variance problem from both sample and distribution perspectives. The main contributions of this work are summarized as follows.

1) The proposed data fusion network called HI^{2}D^{2}FNet for hyperspectral and LiDAR classification is capable of exploring the inherent modality connection from the intrinsic perspective to decrease the interference of their heterogeneous variance. It preliminarily exploits the potential of HIID for multimodal RS data fusion by considering its core mechanism in the whole fusion network. To the best of our knowledge, it is the first work that applies the core mechanism of HIID for hyperspectral and LiDAR FJC to enhance the fusion interpretability.
2) HI^{2}D^{2}FNet presents multiple unique cross-modal constraints in terms of prior RS properties to further enhance the decomposition and fusion interpretability. The local correlations in the reflectance and shading components are maintained by introducing HSI and LiDAR local sample gradients. These constraints make the outputs of the decomposition module explainable and meaningful.
3) HI^{2}D^{2}FNet fuses the hyperspectral and LiDAR features from both sample and distribution perspectives, decreasing the interference of the heterogeneous variance induced by the RS modality inconsistency. Its decomposition and fusion modules are optimized in a novel alternative optimization pattern, trying to achieve a dynamic balance between both the modules for better components and greater classification performance simultaneously.

Section II provides some related works. Section III describes HI^{2}D^{2}FNet in detail. Section IV presents the detailed experimental results and analysis. Section V is the conclusion.

II. Related Work

A. Hyperspectral and LiDAR Data FJC

The complementary property between HSI and LiDAR data has been investigated and applied in numerous RS-related scenarios, including vegetation monitoring, forest management, defense security, etc. LiDAR data compensate for the lack of specific on-ground object elevations in HSIs. Extensive research has been devoted to increasing the modality fusion performance for subsequent joint classification tasks. To exploit multisource information efficiently for Earth observation, Xu et al. [33] designed a novel CNN-based framework with dual streams (DualCNN). Features were extracted from HSI and LiDAR data by one two-tunnel CNN and one CNN with cascade blocks. Afterward, a cascade network was chosen for multisource feature fusion. Xue et al. [34] presented a self-calibrated (SC) convolution-based multiscale DL network (MSNetSC) for hyperspectral and LiDAR joint classification. They combined a hierarchical residual structure with SC convolution to simulate the latent multimodal structure. Li et al. [35] presented an effective morphological convolution and attention calibration network (MCAC) to capture the consistency between HSI and LiDAR data. Zhang et al. [36] provided a multimodal transformer network to capture both the specific and shared RS characteristics. Xu et al. [37] designed a dual-branch dynamic modulation network for underused RS modalities. Yao et al. [38] found that most existing techniques have been designed with exclusive reliance on labeling, which can be limited due to costly production. Thus, they extracted the modality-specific and mutual latent representations in the generated common subspace for accurate fusion results. Furthermore, some techniques [39], [40] construct vision transformers to fuse these two RS modalities. They attempted to improve the basic transformer structure to enhance the cross-modal high-level feature extraction ability.

B. HIID

HIID has been verified to be an effective tool for image enhancement and shadow removal recently. It aims to decompose one HSI into one reflectance component $\boldsymbol{R}$ and one shading component $\boldsymbol{S}$ from the intrinsic perspective. Notably, $\boldsymbol{R}$ and $\boldsymbol{S}$ reflect the detailed object materials and illumination, respectively. In the existing HIID works, $\boldsymbol{S}$ is always removed directly for intraclass consistency and stability. We aim to preliminarily exploit the potential of HIID for multimodal RS data fusion by exploring the inherent connection between $\boldsymbol{S}$ and the LiDAR data with respect to geometry information.

Formally, the HSI is represented as $\boldsymbol{H} \in \mathbb{R}^{m\times n\times b}$, and the basic optimization function of HIID can be represented as

$$\boldsymbol{H}(\lambda) = \boldsymbol{R}(\lambda) \circ \boldsymbol{S}(\lambda) \tag{1}$$

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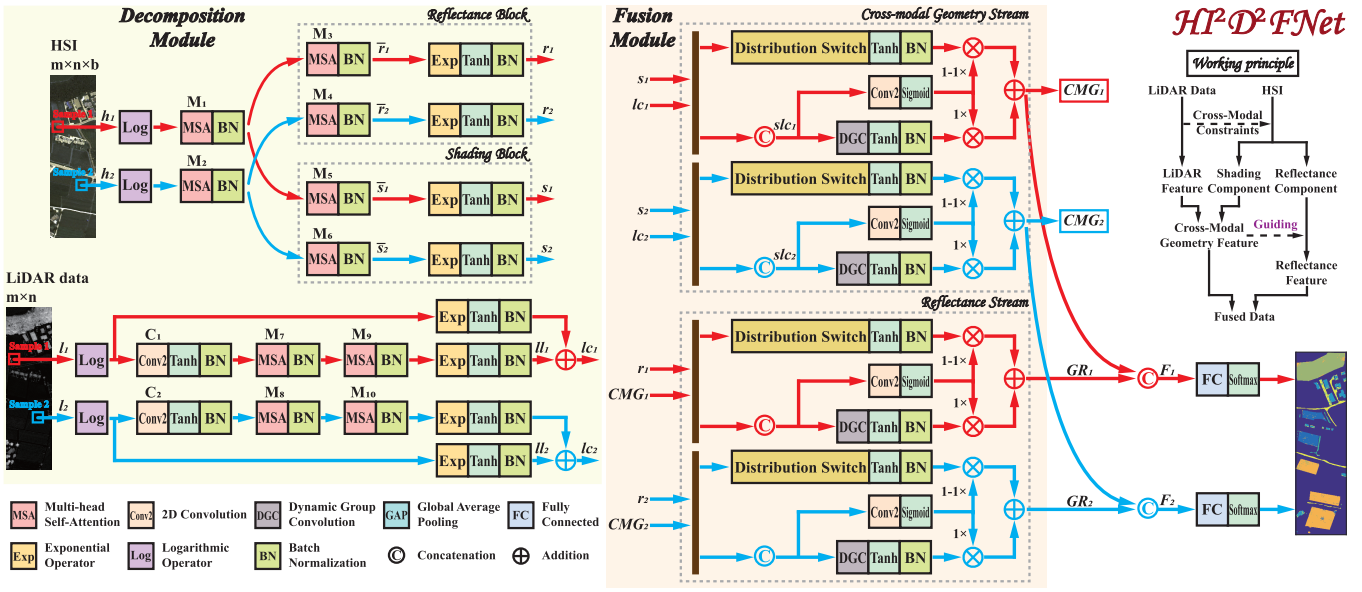

Fig. 2. Scheme of HI^{2}D^{2}FNet for hyperspectral and LiDAR classification. It consists of two separate modules, including the decomposition and fusion modules. They are trained in an alternative optimization pattern. The decomposition module contains two blocks to generate the reflectance and shading components. The fusion module contains the cross-modal geometry and reflectance streams. Two different samples (red and blue samples) are randomly chosen for calculating local sample gradients. Their corresponding structures are also marked in red and blue, respectively. The cross-modal constraints and losses are not shown for clarity. Furthermore, the working principle is also provided for an intuitive comparison with the detailed structure.

where m and n are the length and width of $\boldsymbol{H}$, respectively, b is the channel number, and $\circ$ is the Hadamard product. Both the components vary with the detailed wavelength λ. Apparently, (1) is intractable and complicated to be optimized. Thus, it is rewritten as

$$\log\{\boldsymbol{H}(\lambda)\} = \log\{\boldsymbol{R}(\lambda)\} + \log\{\boldsymbol{S}(\lambda)\}. \tag{2}$$

Considering (2) is severely ill-posed, multiple energy constraints on components are provided with respect to HSI properties. For example, Jin et al. [31] presented dual priors into (2) to reduce the decomposition uncertainty. In view of the low resolution of HSIs, Gu et al. [32] used panchromatic images with high resolution to enhance the spatial details in components. Gao et al. [41] applied HIID in change detection tasks by introducing temporal–spatial energy constraints.

In this article, the core mechanism of HIID is considered in HI^{2}D^{2}FNet to enhance the RS modality fusion interpretability. The detailed feasibilities are listed as follows. First, the HIID-based decomposition module extracts cross-modal geometry features for generating $\boldsymbol{R}$ and $\boldsymbol{S}$ simultaneously. The horizontal geometry information in $\boldsymbol{S}$ describes the object arrangements along the horizontal directions. The vertical geometry information in the LiDAR data records the detailed object elevations along the vertical direction. Considering them together is beneficial to explore their potential relationships with respect to geometry information. The generated cross-modal geometry features with comprehensive geometry information could further guide the reflectance stream optimization. Second, as two crucial modules in HI^{2}D^{2}FNet, the decomposition and fusion modules have explainable and meaningful inputs and outputs, contributing to enhancing the interpretability. Finally, without using the existing HIID methods, HI^{2}D^{2}FNet presents multiple novel cross-modal constraints in terms of prior RS properties, further enhancing this interpretability.

III. Proposed Method

A. Motivation

As shown in Section II-B, classic HIID decomposes one single HSI into one reflectance component and one shading component. The former reflects the object materials, which are invariant to illumination. The latter is determined by the illumination in specific RS scenarios, e.g., shading, sky-lighting, and street-lighting. Furthermore, the shading component also contains abundant horizontal geometry information, describing the object arrangements along the horizontal directions. Since the detailed object elevations are not included in this component, its geometry information is regarded as "horizontal" in this article. Conversely, the LiDAR data contain vertical geometry information and record the detailed object elevations along the vertical direction. On the whole, considering them together contributes to exploring their potential relationships for information integrity with respect to geometry information. This is the chief motivation for introducing the core mechanism of HIID into HI^{2}D^{2}FNet. Subsequently, the obtained cross-modal geometry features are capable of guiding the reflectance stream optimization and bridging the gap between RS modalities.

The intermediate features in most existing DL techniques are unexplainable and meaningless visually. They are just used for extracting discriminant information by black-box networks. To enhance the fusion interpretability, HI^{2}D^{2}FNet constructs the decomposition and fusion modules for explainable and meaningful intermediate features. Both the modules are optimized in an alternative optimization pattern. To further enhance the decomposition and fusion interpretability, HI^{2}D^{2}FNet presents multiple cross-modal constraints in terms

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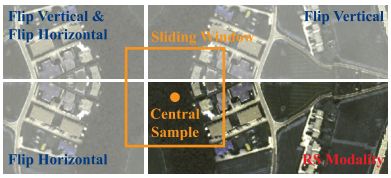

Fig. 3. Mirror symmetry strategy for edge samples in HSI and LiDAR data.

of prior RS properties. Considering the weak and limited application of HIID in multimodal data fusion, our work preliminarily exploits its potential for HSI and LiDAR data fusion.

B. Prepossessing Procedure

The scheme and working principle are shown in Fig. 2. The HSI and LiDAR data are denoted as $\boldsymbol{H} \in \mathbb{R}^{m\times n\times b}$ and $\boldsymbol{L} \in \mathbb{R}^{m\times n}$, respectively. A $w \times w$ sliding window is used to partition $\boldsymbol{H}$ and $\boldsymbol{L}$ into small patches $\boldsymbol{H_p} \in \mathbb{R}^{w\times w\times b}$ and $\boldsymbol{L_p} \in \mathbb{R}^{w\times w}$. The label of the central sample is regarded as the label of both the patches. For edge samples, the mirror symmetry strategy in Fig. 3 avoids the spectral–spatial information loss. Thus, the patch numbers of both the modalities are $m \times n$. All the patches are normalized into $[-1, 1]$ and randomly divided into the training and testing sets subsequently.

C. Decomposition Module

The decomposition module consists of three blocks, including the HSI reflectance block, the HSI shading block, and the LiDAR block. Considering that HI²D²FNet needs to calculate local sample gradients to constrain the decomposition module, two different samples, including their HSI patches (i.e., $\boldsymbol{h_1}$ and $\boldsymbol{h_2}$) and LiDAR patches (i.e., $\boldsymbol{l_1}$ and $\boldsymbol{l_2}$) are randomly chosen as inputs. In Fig. 2, their corresponding network structures are marked in red and blue for a clear explanation. Equation (2) shows the HSI should be transferred into the logarithmic space in the basic HIID optimization function. Thus, HI²D²FNet uses the logarithmic operator $\log(\cdot)$ to transfer all the patches initially, meaning that the generated components should also be transferred by the exponential operator $\exp(\cdot)$. Honestly, directly considering the intractable function in (1) is feasible for DL networks. The superiority of following the optimization strategy in 2 is proven in Section IV-D3.

1) HSI Reflectance and Shading Blocks: Multihead self-attention (MSA) [42] has been verified to have excellent performance in terms of capturing latent features from high-dimensional samples. Based on its superior feature extraction ability, it is capable of highlighting relatively significant segments adaptively and occupying a crucial position in RS interpretation tasks. Inspired by it, the HSI reflectance and shading blocks are constructed for high-level HSI features.

As shown in Fig. 2, two Tanh-function-activated MSA layers M_1 and M_2 are applied for extracting HSI features adaptively from both the samples. Their input shapes are both $[BS, w, w, b]$, where BS is the batch size. Initially, these inputs are reshaped into $[BS, w \times w, b]$ for MSA layers. Since MSA layers partition heads along the spectral dimension, b should be divisible by their head numbers. Then, the outputs are reshaped into $[BS, w, w, b]$ for the following layers. To introduce the core mechanism of HIID into this network, HI²D²FNet uses totally four MSA layers for generating the reflectance and shading components for both the samples. Specifically, M_3 and M_4 belong to the reflectance block, and M_5 and M_6 belong to the shading block. Their structures are consistent with M_1 and M_2. As mentioned above, $\bar{\boldsymbol{r}}_1$, $\bar{\boldsymbol{r}}_2$, $\bar{\boldsymbol{s}}_1$, and $\bar{\boldsymbol{s}}_2$ are still in the logarithmic space. Thus, they are all transferred by $\exp(\cdot)$. Formally, the final outputs of the reflectance and shading blocks, i.e., $\boldsymbol{r}_1$, $\boldsymbol{r}_2$, $\boldsymbol{s}_1$, and $\boldsymbol{s}_2$, are obtained as

$$\boldsymbol{O}_i = \mathrm{Tanh}\{\exp(\mathrm{MSA}[\mathrm{MSA}[\log(\boldsymbol{h}_i)]])\} \tag{3}$$

where $\boldsymbol{O} \in \{\boldsymbol{r}, \boldsymbol{s}\}$, $i \in \{1, 2\}$. Notably, batch normalization (BN) layers are used after all the layers for better optimization performance. Although both the block structures are the same, their detailed functions for decomposition and fusion are absolutely different. These differences are achieved by introducing the following cross-modal constraints and losses.

Equation (2) shows that one HSI can be decomposed into one reflectance component and one shading component by the classic HIID framework. In HI²D²FNet, the corresponding function can be formatted as $\boldsymbol{h}_i = \bar{\boldsymbol{r}}_i + \bar{\boldsymbol{s}}_i$, where $i \in \{1, 2\}$. Therefore, the reconstruction loss L_{REC} is

$$\begin{aligned} L_{\mathrm{REC}} &= L_{\mathrm{REC1}} + L_{\mathrm{REC2}} \\ &= \frac{1}{b}\sum_{\substack{i=1,2\\1\le j\le b}} \left[\mathrm{GAP}(\boldsymbol{h}_i - \bar{\boldsymbol{r}}_i - \bar{\boldsymbol{s}}_i)^j\right]^2 \end{aligned} \tag{4}$$

where GAP is the global average pooling layer. Section III-A proves that the reflectance component contains intrinsic information in terms of object materials. Generally, two reflectance samples should be highly correlated if the following conditions are met simultaneously: 1) they are neighbors in the spatial domain and 2) their spectral curves are similar. Based on these prior RS properties, the first cross-modal constraint loss $L_{\mathrm{HSI}\to\boldsymbol{R}}$, which uses the HSI to constrain the reflectance component, is formatted as

$$\begin{aligned} L_{\mathrm{HSI}\to\boldsymbol{R}} &= \frac{1}{b}\sum_{1\le j\le b}\left\{\boldsymbol{g_H}^j\left[\mathrm{GAP}(\bar{\boldsymbol{r}}_1)^j - \mathrm{GAP}(\bar{\boldsymbol{r}}_2)^j\right]^2\right\} \\ \boldsymbol{g}_{\boldsymbol{H}}^{\boldsymbol{j}} &= \mathrm{Sigmoid}\left\{\left[\mathrm{Dif}(\boldsymbol{h}_1, \boldsymbol{h}_2)^j \times \mathrm{Euc}(\boldsymbol{h}_1, \boldsymbol{h}_2)\right]^{-1}\right\} \\ \mathrm{Dif}(\boldsymbol{h}_1, \boldsymbol{h}_2) &= |\mathrm{GAP}(\boldsymbol{h}_1) - \mathrm{GAP}(\boldsymbol{h}_2)| \end{aligned} \tag{5}$$

where $|\cdot|$ is the absolute value operator, $\boldsymbol{g_H}$ is the improved HSI sample gradient, and $\mathrm{Euc}(\cdot)$ is the Euclidean distance in the spatial domain. For $\boldsymbol{h}_1$ and $\boldsymbol{h}_2$, their corresponding $\boldsymbol{g_H}$ is fixed. If $\boldsymbol{h}_1$ and $\boldsymbol{h}_2$ are neighbors, $\boldsymbol{g_H}$ is bigger/smaller if their spectral curves are more/less similar. Bigger $\boldsymbol{g_H}$s makes it easy to narrow the gap between $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ when minimizing $L_{\mathrm{HSI}\to\boldsymbol{R}}$. If $\boldsymbol{h}_1$ and $\boldsymbol{h}_2$ are not neighbors, $\mathrm{Euc}(\boldsymbol{h}_1, \boldsymbol{h}_2)$ is very big, making $\boldsymbol{g_H}$ quite small (maybe close to 0). Thus, the gap between $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ is hard to be narrowed by just minimizing $L_{\mathrm{HSI}\to\boldsymbol{R}}$. The second cross-modal constraint loss $L_{\mathrm{LiDAR}\to\boldsymbol{S}}$ is provided in Section III-C2.

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Considering $\boldsymbol{R}$ and $\boldsymbol{S}$ have no information overlap and should be absolutely different ideally, the divergence loss L_{DIV} measuring the divergence between components is

$$L_{\mathrm{DIV}} = \frac{1}{b} \sum_{\substack{i=1,2 \\ 1 \le j \le b}} \left\{ \frac{\mathrm{GAP}(\boldsymbol{r_i})^j \times \mathrm{GAP}(\boldsymbol{s_i})^j}{||\mathrm{GAP}(\boldsymbol{r_i})|| \times ||\mathrm{GAP}(\boldsymbol{s_i})||} \right\} \tag{6}$$

where $||\cdot||$ is the Frobenius norm. If the spectral-angle-based L_{DIV} is minimized, the component divergence becomes large.

In general, the HSI reflectance and shading blocks aim to decompose the HSI into two components by considering multiple cross-modal constraints and losses. Although the core mechanism of HIID is relatively simple, great decomposition performance can be obtained by exploring the inherent modality connection in terms of prior RS properties.

2) LiDAR Block: Similar to the HSI blocks, the LiDAR block captures high-level features from the LiDAR data and constrains the component decomposition. Since the LiDAR patch shape is [BS, w, w], two Tanh-function-activated 2-D convolution (Conv2) layers C_1 and C_2 with b 3 × 3 kernels are applied (SAME padding), making the patch shapes same with HSI patches. Subsequently, two sequential MSA layers extract high-level features for each sample. They share the same hyperparameters with other MSA layers. After that, the outputs are transferred by $\exp(\cdot)$ for $\boldsymbol{ll_1}$ and $\boldsymbol{ll_2}$. Since the Conv2 layers would cause the LiDAR information shift inevitably, a cross-layer addition strategy is used for the final outputs $\boldsymbol{lc_1}$ and $\boldsymbol{lc_2}$. Specifically, $\boldsymbol{l_1}$ and $\boldsymbol{l_2}$ are added with all the channels of $\boldsymbol{ll_1}$ and $\boldsymbol{ll_2}$, respectively.

Theoretically, the shading component reflects the detailed illumination and has no connection with the object materials. In this article, we assume that samples with similar elevations always have similar light intensities. This assumption simplifies the whole framework by ignoring the effect of shadows on decomposition performance. The reasons are listed as follows. First, the number of samples near shadow edges is always limited in the whole target region. Only these edge samples may be influenced by shadows. Second, an appropriate amount of inaccurate information tends to increase the robustness of the whole network. 3) Finally, the HSI and LiDAR data suffer from challenges because of the low resolution. Thus, this assumption would not influence the final performance. Hence, two shading samples should be highly correlated if their elevation variation is small. The second cross-modal constraint loss $L_{\mathrm{LiDAR}\to S}$, using the LiDAR data to constrain the shading component, is formatted as

$$\begin{aligned} L_{\mathrm{LiDAR}\to S} &= \frac{1}{b} \sum_{1 \le j \le b} \left\{ g_L \left[\mathrm{GAP}(\bar{s}_1)^j - \mathrm{GAP}(\bar{s}_2)^j \right]^2 \right\} \\ g_L &= \mathrm{Sigmoid}\left\{ [\mathrm{Dif}(\boldsymbol{l_1}, \boldsymbol{l_2}) \times \mathrm{Euc}(\boldsymbol{l_1}, \boldsymbol{l_2})]^{-1} \right\} \\ \mathrm{Dif}(\boldsymbol{l_1}, \boldsymbol{l_2}) &= |\mathrm{GAP}(\boldsymbol{l_1}) - \mathrm{GAP}(\boldsymbol{l_2})| \end{aligned} \tag{7}$$

where g_L is the improved LiDAR sample gradient. If two neighboring samples have similar elevations, the gap between their corresponding shading samples will be narrowed by minimizing $L_{\mathrm{LiDAR}\to S}$.

D. Fusion Module

1) Cross-Modal Geometry Stream: The cross-modal geometry stream aims to fuse the shading component and the LiDAR data together for exploring their potential relationships for information integrity with respect to geometry information. The fusion feasibilities are listed as follows. First, the HSI and LiDAR data are captured over the same target region, describing the same on-ground objects from two complementary perspectives. Thus, the features extracted from the decomposed HSI shading component and the LiDAR data are homologous to some extent. Second, as analyzed in Section III-A, they contain horizontal and vertical geometry information, respectively. Fusing them together is beneficial to describe the detailed geometry structure comprehensively.

Specifically, this stream fuses $\boldsymbol{s_i}$ with $\boldsymbol{lc_i}$ from both sample and distribution perspectives to generate the cross-modal geometry feature CMG_i, where $i \in \{1, 2\}$. For the sample alignment, dynamic group convolution (DGC) layers [43] are considered since they have been verified to be efficient substitutes for classic convolution layers in multimodal fusion tasks. They are capable of capturing the inherent channel connection among modality samples and grouping nonadjacent channels. For $\boldsymbol{s_1}$ and $\boldsymbol{lc_1}$, one concatenation layer merges them along the channel dimension for $\boldsymbol{slc_1} \in \mathbb{R}^{w \times w \times 2b}$ initially. The DGC layer with b 3 × 3 kernels is formatted as

$$\boldsymbol{slc_1^{out1}}(i, j) = \sum_{1 \le x, y \le 3} \left[\boldsymbol{p}(x, y) \circ \boldsymbol{P} \right] \boldsymbol{slc_1}(i + x, j + y) \tag{8}$$

where $\boldsymbol{slc_1^{out1}}$ is the output, $1 \le i, j \le w$, $\boldsymbol{p} \in \mathbb{R}^{3 \times 3 \times b \times 2b}$ is the weight matrix, $\boldsymbol{P} = \hat{\boldsymbol{P}}[1 : b, 1 : 2b]$, $\hat{\boldsymbol{P}} \in \{0, 1\}^{2^K \times 2^K}$, and $K = \log_2(2b)$. $\hat{\boldsymbol{P}}$ is formatted as

$$\begin{aligned} \hat{\boldsymbol{P}} &= \hat{\boldsymbol{P}}_1 \otimes \cdots \otimes \hat{\boldsymbol{P}}_K = \left[z_1 \mathbf{1}_{(2,2)} + (1 - z_1) \boldsymbol{I}_{(2,2)} \right] \\ &\quad \otimes \cdots \otimes \left[z_K \mathbf{1}_{(2,2)} + (1 - z_K) \boldsymbol{I}_{(2,2)} \right] \\ z_1 &= \mathrm{sign}\left(\hat{z}_1\right), \ldots, z_K = \mathrm{sign}\left(\hat{z}_K\right) \end{aligned} \tag{9}$$

where $\mathbf{1}_{(2,2)}$ is the 2 × 2 matrix with all 1 s, $\boldsymbol{I}_{(2,2)}$ is the 2 × 2 identity matrix, $\otimes$ is the Kronecker operator, $\mathrm{sign}(\cdot)$ is the sign function, and $\hat{z}$s are trainable parameters. For the distribution alignment, a distribution switch layer is presented as

$$\begin{aligned} \boldsymbol{slc_1^{out2}} &= (\boldsymbol{ex_1} + \boldsymbol{ex_2})/2 \\ \boldsymbol{ex_1} &= \sigma(\boldsymbol{lc_1}) \circ \left[(\boldsymbol{s_1} - \mu(\boldsymbol{s_1}))/\sigma(\boldsymbol{s_1}) \right] + \mu(\boldsymbol{lc_1}) \\ \boldsymbol{ex_2} &= \sigma(\boldsymbol{s_1}) \circ \left[(\boldsymbol{lc_1} - \mu(\boldsymbol{lc_1}))/\sigma(\boldsymbol{lc_1}) \right] + \mu(\boldsymbol{s_1}) \end{aligned} \tag{10}$$

where $\mu(\cdot)$ and $\sigma(\cdot)$ are the average and standard deviation, respectively. This layer increases the consistency between $\boldsymbol{s_1}$ and $\boldsymbol{lc_1}$ by aligning both the distributions. Finally, an adaptive tradeoff parameter matrix $\boldsymbol{\delta_1} \in \mathbb{R}^{w \times w \times b}$ is learned by one Sigmoid-function-activated Conv2 layer as $\boldsymbol{\delta_1} = \mathrm{Sigmoid}\{\mathrm{Conv2}[\boldsymbol{slc_1}]\}$. It balances $\boldsymbol{slc_1^{out1}}$ and $\boldsymbol{slc_1^{out2}}$ as

$$\mathrm{CMG_1} = \boldsymbol{\delta_1} \circ \boldsymbol{slc_1^{out1}} + (1 - \boldsymbol{\delta_1}) \circ \boldsymbol{slc_1^{out2}}. \tag{11}$$

Similarly, for $\boldsymbol{s_2}$ and $\boldsymbol{lc_2}$, CMG_2 is obtained.

Some fusion techniques meet the information misalignment challenge. Specifically, they tend to distort and lose unique data characteristics inevitably. Compared with them,

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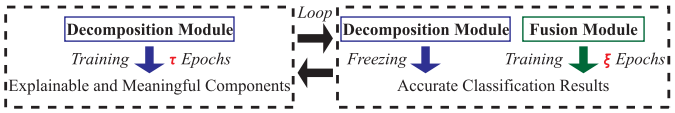

Fig. 4. Scheme of the alternative optimization pattern.

HI²D²FNet is capable of generating acceptable results in misaligning scenarios because it fuses RS modalities from both sample and distribution perspectives. The MSA layers in the decomposition module further enhance this robustness. The classifiers in this network contribute to retaining useful cross-modal information and removing interference and redundant information for better fusion performance. Furthermore, the fusion feasibilities analyzed above for HSI and LiDAR data prove that these modalities are homologous to some extent.

2) Reflectance Stream: The reflectance stream aims to guide its optimization by the generated cross-modal geometry features. Thus, $\mathrm{CMG_1}$ and $\mathrm{CMG_2}$ are delivered into this stream. As shown in Fig. 2, the network structures of the cross-modal geometry and reflectance streams are the same. They both fuse their inputs from sample and distribution perspectives. The input couples for both the samples in the reflectance stream are $\langle \boldsymbol{r_1}, \mathrm{CMG_1} \rangle$ and $\langle \boldsymbol{r_2}, \mathrm{CMG_2} \rangle$, respectively. The outputs are $\boldsymbol{GR_1}$ and $\boldsymbol{GR_2}$ for both the samples.

3) Cross-Modal Feature Fusion and Classification: The cross-modal geometry features are concatenated with the outputs of the reflectance stream for feature fusion and classification. Subsequently, the fused features $\boldsymbol{F_1}$ and $\boldsymbol{F_2}$ for both the samples are classified by Softmax-function-activated fully connected (FC) layers. The classification loss L_{CLS} is

$$L_{\mathrm{CLS}} = -\sum_{\substack{i=1,2\\1\le j\le b}} \boldsymbol{c}_i^j \log\left(\mathrm{Flatten}\left[\boldsymbol{F}_i^j\right]\right) - \sum_{\substack{i=1,2\\1\le j\le b}} \left(1-\boldsymbol{c}_i^j\right) \log\left(1-\mathrm{Flatten}\left[\boldsymbol{F}_i^j\right]\right) \quad (12)$$

where $\boldsymbol{c_i}$ is the one-hot label of $\boldsymbol{F_i}$, and Flatten$[\cdot]$ is the flatten layer. The reason why using two different classifiers is that these two samples with different labels are randomly selected. Existing works have proved that improving the consistency between feature segments is beneficial to great classification performance [44]. Thus, the consistency loss L_{CON} is

$$L_{\mathrm{CON}} = \frac{1}{b}\sum_{\substack{i=1,2\\1\le j\le b}} \left\{1 - \frac{\mathrm{GAP}(\mathrm{CMG}_i)^j \times \mathrm{GAP}(\boldsymbol{GR}_i)^j}{||\mathrm{GAP}(\mathrm{CMG}_i)|| \times ||\mathrm{GAP}(\boldsymbol{GR}_i)||}\right\}. \quad (13)$$

The consistency between CMG_i and $\boldsymbol{GR_i}$ is improved by minimizing L_{CON}.

E. Alternative Optimization Pattern and Optimization

The decomposition module aims to generate diverse components by introducing prior RS properties since the components should have no information overlap ideally. The fusion module aims to increase the consistency between CMG and $\boldsymbol{GR}$ for tackling the heterogeneous variance problem. Apparently, their optimization directions are discrepant. In Fig. 4, an alternative optimization pattern is proposed to train both the modules successively. Specifically, the optimization loop is: 1) training the decomposition module for explainable and meaningful components in τ epochs and 2) freezing the decomposition module and training the fusion module for accurate classification results in ξ epochs.

HI²D²FNet repeats these two steps alternatively and stops the training procedure when the loss value converges. Training either module could put convergence pressure on the other module to increase its generation/fusion ability. This pattern attempts to achieve a dynamic balance between both the modules for better decomposed components and greater classification performance simultaneously. There are two reasons why all the losses are not combined and used to train the whole network simultaneously: 1) different types of losses with discrepant optimization directions could better harness their strengths when they are used separately and 2) a relatively large network with multiple constraints and losses tend to weaken and neglect the contribution of the single loss inevitably. Thus, when training the decomposition and fusion modules separately, their loss functions L_D and L_F are

$$\begin{aligned} L_D &= L_{\mathrm{REC}} + L_{\mathrm{HSI}\to\boldsymbol{R}} + L_{\mathrm{LiDAR}\to\boldsymbol{S}} + L_{\mathrm{DIV}} \\ L_F &= \alpha L_{\mathrm{CLS}} + L_{\mathrm{CON}} \end{aligned} \quad (14)$$

where α is the classification-related loss weight. To reduce the hyperparameter number, other loss weights are set as 1.

The computing complexity is $\mathcal{O}(w^2b^2)$, where w is the sliding window side, and b is the HSI channel number. It illustrates that the computing time is mainly affected by w and b. Practically, the training time is influenced by several factors, including epoch number and loss function convergence.

TABLE I
DETAILED SAMPLE NUMBERS IN THE MUUFL DATASET

Class	Land-cover Type	Training	Testing	Total
1	Trees	232	23014	23246
2	Mostly Grass	42	4228	4270
3	Mixed Ground Surface	69	6813	6882
4	Dirt and Sand	18	1808	1826
5	Road	67	6620	6687
6	Water	5	461	466
7	Building Shadow	22	2211	2233
8	Building	62	6178	6240
9	Sidewalk	14	1371	1385
10	Yellow Curb	2	181	183
11	Cloth Panels	3	266	269

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Dataset Description

1) Muufl Gulfport Dataset: The Muufl gulfport (MUUFL) dataset was acquired over the University of Southern Mississippi Gulf Park Campus, MS, USA, in 2010. The HSI was acquired by the CASI-1500 sensor with 64 spectral channels ranging from 375 to 1050 nm. They both contain 325 × 220 samples with 1-m spatial resolution. There are totally 11 land-cover categories in this dataset. Their detailed

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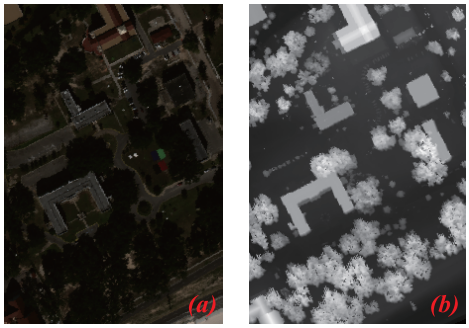

Fig. 5. Visualization of the MUUFL dataset. (a) Pseudocolor image of the HSI. (b) Gray image of the LiDAR DSM data.

TABLE II
DETAILED SAMPLE NUMBERS IN THE TRENTO DATASET

Class	Land-cover Type	Training	Testing	Total
1	Apple Trees	40	3994	4034
2	Buildings	29	2874	2903
3	Ground	5	474	479
4	Wood	91	9032	9123
5	Vineyard	105	10396	10501
6	Roads	32	3142	3174

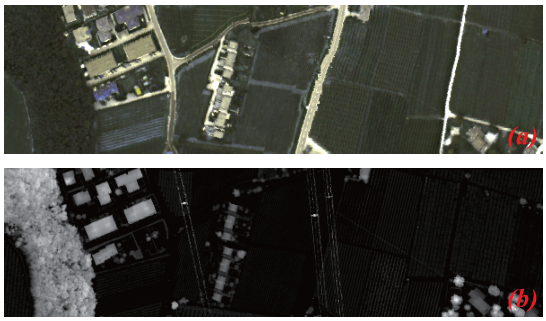

Fig. 6. Visualization of the Trento dataset. (a) Pseudocolor image of the HSI. (b) Gray image of the LiDAR DSM data.

sample numbers are shown in Table I. One pseudocolor image of the HSI and one gray image of the LiDAR DSM data of this dataset are shown in Fig. 5.

2) Trento Dataset: The Trento dataset was acquired over a rural region in the south of Trento, Italy. It contains one HSI and one LiDAR DSM data. The former was captured by the AISA Eagle sensor with 63 spectral channels ranging from 402.89 to 989.09 nm. The latter was captured by the Optech ALTM 3100EA sensor. Both of them have 166 × 600 samples. Their spatial resolution is 1 m. The detailed sample numbers in all the categories are shown in Table II. One pseudocolor image of the HSI and one gray image of the LiDAR DSM data are shown in Fig. 6.

3) Houston Dataset: The Houston dataset was provided for the 2018 IEEE GRSS Data Fusion Contest. It was acquired by the National Center for Airborne Laser Mapping (NCALM) over the campus of the Houston University in 2017. It contains one HSI with 48 spectral channels ranging from 380 to 1050 nm and one LiDAR data. Their spatial resolutions are 1 m and 50 cm, respectively. An interval sampling strategy

TABLE III
DETAILED SAMPLE NUMBERS IN THE HOUSTON SUBSCENE

Class	Land-cover Type	Training	Testing	Total
1	Healthy Grass	14	1439	1453
2	Stressed Grass	51	5020	5071
3	Evergreen Trees	64	6371	6435
4	Deciduous Trees	30	2925	2955
5	Residential Buildings	81	7982	8063
6	Non-Residential Buildings	482	47782	48264
7	Roads	70	6969	7039
8	Sidewalks	125	12372	12497
9	Major Thoroughfares	179	17722	17901
10	Paved Parking Lots	42	4114	4156
11	Cars	11	1090	1101
12	Stadium Seats	68	6756	6824

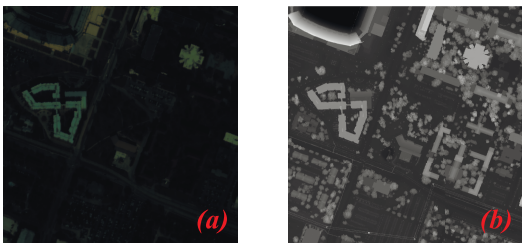

Fig. 7. Visualization of the Houston subscene. (a) Pseudocolor image of the HSI. (b) Gray image of the LiDAR DSM data.

was used to unify the modality shapes. Considering the large dataset scale, one partitioned subscene with 601 × 596 samples (i.e., UH_NAD83_272652_3289689) was selected in this article. The sample numbers in all the categories are provided in Table III. The dataset visualizations are shown in Fig. 7.

Notably, in all the three datasets, the categories were directly removed if their sample ratios are less than 2‰. The sample ratio is the ratio of the category sample number to the total sample number in the dataset. Furthermore, totally 1% samples per category were randomly chosen for training. The remaining 99% samples were used for testing.

B. Experimental Setting

The experiments were conducted on a PC (16-GB RAM) with Intel Core i7-8750 CPU and the DL workstation with NVIDIA GeForce RTX 2080 Ti GPU. All the codes were implemented in the Keras framework with TensorFlow backend using Python 3.7.0 (Spyder). To evaluate the FJC performance quantitatively, totally six classification metrics were selected, including the accuracy per category, overall accuracy (OA), average accuracy (AA), kappa coefficient (κ), precision (PR), and F1-score (F1). Furthermore, the generated classification maps were provided for evaluation qualitatively. The adaptive moment estimation (Adam) was chosen for updating parameters in all the layers. Its parameters were set as: $\beta_1 = 0.9$, $\beta_2 = 0.99$, and $\epsilon = 10^{-8}$ and the decay was 0. The batch size was selected within $\{128, 256, 512, 1024\}$. The learning rate was set as 5×10^{-3}. The loss weight α was set as 1. The sliding window side w was set as 5. The head numbers for the three datasets were set as 4, 3,

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TABLE IV
CLASSIFICATION RESULTS ON THE MUUFL DATASET

Class	Early	Middle	Late	ENDnet	CNN$_H$	CNN$_L$	IPCNN	DualCNN	MVHN	MSNetSC	EMFnet	CPCNN	MCAC	LIIT	Our
1	94.42	94.36	92.64	93.75	96.57	84.27	96.15	96.44	96.58	96.20	96.90	**98.77**	97.74	96.72	96.95
2	51.40	64.10	51.49	71.14	63.55	33.96	60.00	67.24	66.75	74.01	71.81	68.90	69.70	66.91	**77.51**
3	82.03	65.60	74.08	66.74	76.12	35.40	79.83	80.74	73.36	81.08	81.56	77.59	83.21	79.14	**84.15**
4	69.03	73.34	80.92	47.51	59.02	4.48	78.76	81.42	56.80	**91.54**	74.61	72.95	79.76	75.88	89.88
5	93.23	86.72	91.10	88.94	93.04	62.81	92.52	93.26	83.08	92.66	91.50	91.96	**93.69**	91.47	92.60
6	72.89	83.73	73.97	82.00	45.55	74.40	74.40	75.05	41.00	95.66	81.56	4.34	59.22	89.15	**96.10**
7	47.04	64.18	62.96	67.62	62.60	15.65	74.90	71.46	65.08	81.28	75.03	41.20	70.87	69.74	**83.49**
8	86.73	90.84	88.94	91.19	95.03	76.17	83.38	96.46	93.40	**97.17**	95.39	92.76	95.00	95.60	95.95
9	57.91	50.40	**61.05**	47.78	41.58	2.41	45.08	39.61	17.29	34.21	34.65	28.15	45.15	32.53	31.00
10	24.86	**28.73**	25.41	15.47	0	0.55	5.52	4.97	2.21	0	3.31	0	7.18	25.41	1.10
11	41.73	59.40	56.39	**87.97**	66.92	0	42.11	48.87	42.11	40.98	60.90	38.35	45.11	45.11	50.00
OA (%)	83.90	83.31	83.35	83.35	85.67	61.92	85.68	87.96	83.40	89.18	88.18	85.58	88.83	87.26	**90.03**
AA (%)	65.57	69.22	69.00	69.10	63.63	35.46	66.60	68.68	57.97	71.34	69.75	55.91	67.88	69.79	**72.61**
κ (÷100)	78.60	77.97	78.04	78.00	80.92	49.11	80.89	84.01	77.86	85.68	84.29	80.36	85.10	83.08	**86.80**
PR (%)	75.19	68.96	73.87	68.61	72.55	37.56	73.12	76.98	65.35	76.39	76.22	71.65	**77.45**	72.95	**77.45**
F1 (÷100)	68.40	68.43	70.55	68.71	67.02	35.21	68.76	71.67	60.62	72.50	72.13	60.05	71.62	70.75	**74.09**

TABLE V
CLASSIFICATION RESULTS ON THE TRENTO DATASET

Class	Early	Middle	Late	ENDnet	CNN$_H$	CNN$_L$	IPCNN	DualCNN	MVHN	MSNetSC	EMFnet	CPCNN	MCAC	LIIT	Our
1	66.85	64.37	82.37	69.23	99.10	89.26	90.49	**99.62**	97.45	92.21	97.25	99.20	93.16	96.19	**99.62**
2	97.63	95.09	97.22	95.06	89.98	79.75	97.95	98.54	92.14	97.77	96.87	86.99	**99.10**	94.33	98.57
3	91.14	93.88	86.29	78.69	94.73	13.71	91.14	92.62	87.76	90.08	94.30	47.68	91.98	**98.10**	93.88
4	99.25	98.77	99.07	98.48	99.35	92.36	99.99	**100.00**	99.68	98.49	99.97	**100.00**	**100.00**	99.79	**100.00**
5	90.04	91.84	88.27	87.17	99.84	98.02	96.94	99.95	98.03	96.50	99.10	99.12	**100.00**	99.90	99.93
6	89.88	90.80	87.62	91.15	77.59	76.89	92.01	90.10	87.11	92.17	95.51	97.20	92.78	**97.58**	94.56
OA (%)	90.46	90.50	91.51	89.23	96.23	89.83	96.49	98.64	96.57	96.10	98.45	97.21	98.11	98.57	**99.12**
AA (%)	89.13	89.13	90.14	86.63	93.43	75.00	94.75	96.81	93.69	94.54	97.17	88.36	96.17	97.65	**97.76**
κ (÷100)	87.22	87.24	88.71	85.63	94.96	86.33	95.31	98.18	95.43	94.80	97.93	96.27	97.48	98.09	**98.83**
PR (%)	88.77	90.52	88.71	88.16	91.74	81.87	93.47	96.42	93.84	92.81	96.13	96.04	98.01	95.68	**98.07**
F1 (÷100)	88.85	89.72	89.26	87.36	92.31	76.40	94.02	96.55	93.74	93.60	96.61	90.82	97.03	96.52	**97.90**

and 4, respectively. The selection of their optimal values was analyzed in Sections IV-E and IV-F. All the classification procedures were repeated ten times to remove the effect of random selection. The training process would be stopped if all the network parameters converged.

Totally, 14 state-of-the-art techniques were chosen for comparison, including three baselines [20] (i.e., early fusion, middle fusion, and late fusion), ENDnet [20], two CNN-based networks [26] (i.e., CNN$_H$ and CNN$_L$), interleaving perception convolutional neural network (IPCNN) [45], DualCNN [33], MVHN [12], MSNetSC [34], EMFnet [27], CPCNN [26], MCAC [35], and local information interaction transformer (LIIT) [10]. Their structures and settings were fine-tuned for a fair comparison. Their explanations are given as follows.

1) *ENDnet* has several FC layers and reconstructs the multimodal inputs without using any spatial information.
2) *CNN$_H$* has three CNN layers and one Softmax-activated FC layer to classify HSI samples.
3) *CNN$_L$* shares the same structure with CNN$_H$ to classify samples in the LiDAR data.
4) *IPCNN* has one unsupervised bidirectional autoencoder to reconstruct HSI and LiDAR samples. It introduces several constraints for accurate fusion.
5) *DualCNN* has one dual-stream cascade CNN to capture the connection between HSI and LiDAR samples.
6) *MVHN* has one stacking-mechanism-based CNN framework. Its structure is quite similar to the classic residual network (ResNet). The extracted principal components are processed by the Gabor filters initially.
7) *MSNetSC* has one SC CNN-based network and captures multimodal features from multiple receptive fields. It has one self-attention module to calibrate the optimized network weights.
8) *EMFnet* has one enhanced modality fusion network and one novel feature fusion module.
9) *CPCNN* has one CPCNN-based network to capture feature-level and decision-level features, simultaneously.
10) *MCAC* has one effective morphological convolution and attention calibration network for HSI and LiDAR data FJC.
11) *LIIT* has one MSA-based LIIT for fusing cross-modal RS information.

C. Experimental Results and Comparison

The classification results on three datasets are provided in Tables IV–VI, respectively. The best results per row are marked in bold. Pixelwise methods are not capable of capturing the spatial information of samples. Thus, these three baselines and ENDnet perform worse than other patchwise methods. CNN$_H$ and CNN$_L$ cannot generate results with high

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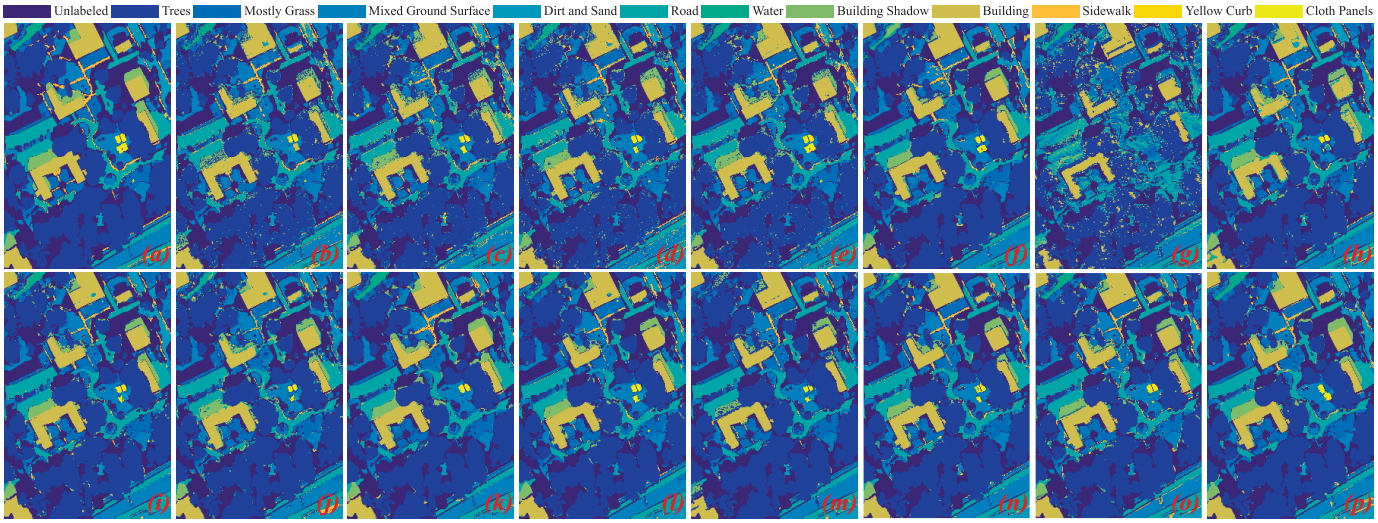

Fig. 8. (a) Ground truth and (b)–(p) classification maps of state-of-the-art techniques on the MUUFL dataset, including (b) EF, (c) MF, (d) LF, (e) ENDnet, (f) CNN_H, (g) CNN_L, (h) IPCNN, (i) DualCNN, (j) MVHN, (k) MSNetSC, (l) EMFnet, (m) CPCNN, (n) MCAC, (o) LIIT, and (p) $\text{HI}^2\text{D}^2\text{FNet}$.

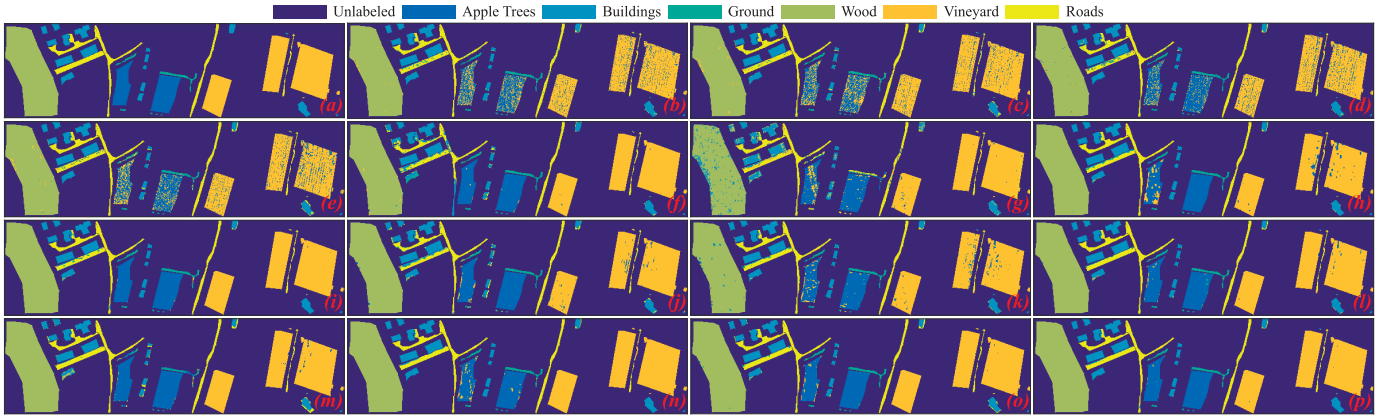

Fig. 9. (a) Ground truth and (b)–(p) classification maps of state-of-the-art techniques on the Trento dataset, including (b) EF, (c) MF, (d) LF, (e) ENDnet, (f) CNN_H, (g) CNN_L, (h) IPCNN, (i) DualCNN, (j) MVHN, (k) MSNetSC, (l) EMFnet, (m) CPCNN, (n) MCAC, (o) LIIT, and (p) $\text{HI}^2\text{D}^2\text{FNet}$.

TABLE VI
CLASSIFICATION RESULTS ON THE HOUSTON DATASET

Class	DualCNN	MVHN	MSNetSC	EMFnet	CPCNN	MCAC	LIIT	Our
1	76.86	75.82	78.39	78.74	**81.58**	60.81	50.73	74.70
2	80.18	**83.17**	77.03	79.48	64.90	78.41	74.96	78.55
3	93.99	75.53	85.73	87.47	**96.08**	89.70	86.38	86.89
4	76.31	78.36	82.43	73.23	84.24	62.19	84.68	**86.97**
5	81.96	79.27	76.35	75.28	88.76	85.40	**95.16**	88.51
6	94.97	96.17	93.27	**96.57**	93.26	95.02	93.89	93.42
7	26.09	38.79	36.65	32.97	21.84	36.05	**44.08**	36.10
8	67.50	65.71	68.63	**71.57**	61.09	60.04	58.97	70.44
9	86.38	76.83	72.23	79.35	**88.64**	82.50	78.15	82.46
10	84.71	87.68	86.63	85.51	**90.18**	83.13	89.57	83.01
11	**71.19**	57.52	29.27	67.89	57.06	59.54	54.40	57.34
12	97.29	93.18	94.46	96.60	**98.90**	97.57	96.05	95.54
OA (%)	83.75	82.65	81.00	83.78	83.35	82.78	82.89	**83.91**
AA (%)	75.16	75.67	73.42	77.06	77.21	74.20	75.59	**77.83**
κ (÷100)	79.37	78.00	76.05	79.38	79.09	78.26	78.48	**79.78**
PR (%)	78.76	75.89	75.03	**79.09**	76.51	77.28	78.54	78.83
F1 (÷100)	75.89	75.42	73.15	77.81	75.88	75.43	76.46	**78.00**

accuracies since they only use one single RS modality. The results illustrate that HSI and LiDAR data are complementary in terms of discriminant information. On the whole, the patchwise methods could generate relatively greater results. Specifically, $\text{HI}^2\text{D}^2\text{FNet}$ outperforms other methods and increases the OA by 0.85% than MSNetSC, which is the best patchwise method. Although EMFnet outperforms $\text{HI}^2\text{D}^2\text{FNet}$ in terms of PR on the Houston dataset, all the results could clearly illustrate the effectiveness of $\text{HI}^2\text{D}^2\text{FNet}$ in terms of all the metrics comprehensively and verify the contributions of the modules in this network. Although LIIT also considers MSA layers for feature extraction, it works worse than $\text{HI}^2\text{D}^2\text{FNet}$, showing the effectiveness of the core mechanism of HIID in this article. Although the tenth class (i.e., *Yellow Curb*) in the MUUFL dataset contains 183 samples, this category is not removed for a fair comparison since its sample ratio is 2.56‰ (larger than 2‰). Thus, there are two samples for training in this category, increasing the optimization difficulty. In this setting, most methods (except CNN_H, MSNetSC, and CPCNN) could extract features from this category, proving their feature extraction abilities. Notably, pixelwise methods perform better than patchwise methods when classifying samples from small categories. The reason is that the former could not be interfered with by neighboring samples which do not belong to the same category. For the Houston dataset, pixelwise methods are not selected for simplification.

The classification maps on three datasets are shown in Figs. 8–10, respectively. Although the differences between two maps are hard to be captured when their OAs are similar,

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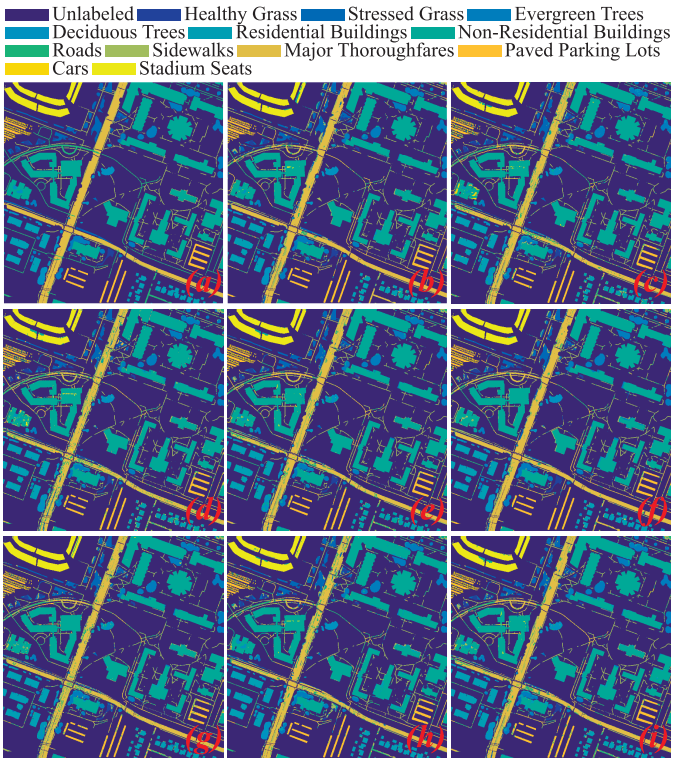

Fig. 10. (a) Ground truth and (b)–(i) classification maps of state-of-the-art techniques on the Houston dataset, including (b) DualCNN, (c) MVHN, (d) MSNetSC, (e) EMFnet, (f) CPCNN, (g) MCAC, (h) LIIT, and (i) HI²D²FNet.

it can still be seen that HI²D²FNet is capable of generating better maps with fewer wrongly classified samples in some categories than other methods. For example, the *mixed ground surface* regions (MUUFL) generally contain fewer scattered noise by HI²D²FNet. Most *apple trees* samples (Trento) and *deciduous trees* samples (Houston) can be correctly classified by HI²D²FNet. Notably, the conclusions drawn from the maps are consistent with those from the aforementioned classification accuracies. The accuracies and maps should be considered together for performance evaluation.

Fig. 11 aims to prove that HI²D²FNet could generate explainable and meaningful components. In Fig. 11, the obtained reflectance and shading components in three optimization scenarios are provided for comparison, including: 1) only using L_{CLS} and removing other losses; 2) using L_{DIV}, L_{CON}, and L_{CLS}, removing all the HIID-mechanism-related losses; and 3) retaining all the losses in HI²D²FNet. Some typical regions are marked by blue squares. It shows that the components in the third scenario are more explainable and meaningful than those in other scenarios. Compared with the HSI and LiDAR data, the reflectance component by HI²D²FNet clearly reflects the land-cover materials in a uniform illumination. The trees and buildings can be accurately identified. The shading component by HI²D²FNet records the illumination, contributing to land-cover classification. Although the interpretability is hard to be evaluated by quantitative metrics, these visualization results at least prove that the core mechanism of HIID in HI²D²FNet contributes to increasing the fusion interpretability.

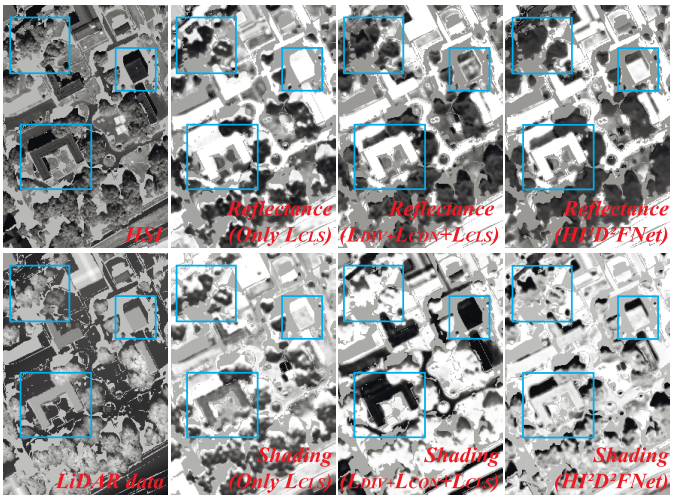

Fig. 11. Visualization of the modalities (i.e., HSI and LiDAR data) and the reflectance and shading components in three optimization scenarios on the MUUFL dataset. Unlabeled samples are removed directly.

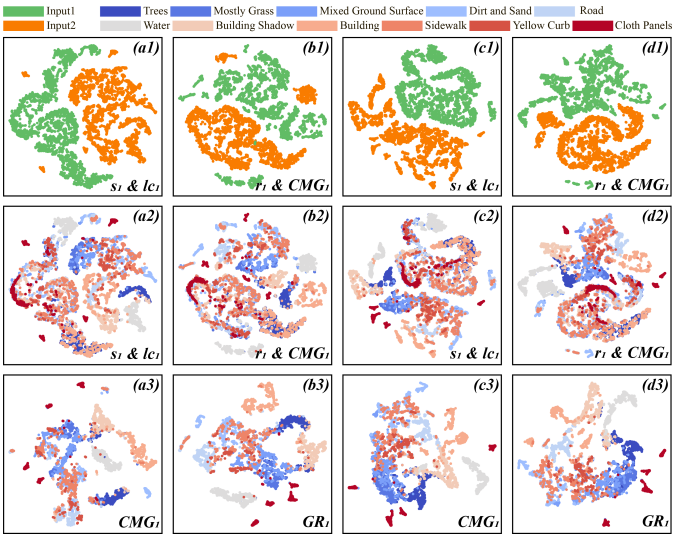

Fig. 12. T-SNE visualization of the inputs and outputs of (a) cross-modal geometry stream and (b) reflectance stream. For comparison, (c) and (d) correspond to the inputs and outputs of both the streams when removing all the alignment-related layers. Two color codings are shown. (1) Colored according to the source. (2) and (3) Colored according to the ground truth.

The t-distributed stochastic neighbor embedding (T-SNE) plots of the inputs and outputs of both the streams in the fusion module are provided in Fig. 12 to evaluate the aligning performance. Different from other classic aligning techniques [46], HI²D²FNet aligns and fuses both the inputs for output generation. Thus, the output and both the inputs contain the same number of samples, making the output sample sources be not tracked. Nevertheless, the aligning performance can be evaluated by category clustering to a certain extent. In Fig. 12, the T-SNE plots include: (a1) s_1 and lc_1 colored according to the source, (a2) s_1 and lc_1 colored according to the ground truth, (a3) CMG_1 colored according to the ground truth, (b1) r_1 and CMG_1 colored according to the source, (b2) r_1 and CMG_1 colored according to the ground truth, and (b3) GR_1 colored according to the ground truth. For comparison, all the alignment-related layers, e.g., DGC

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TABLE VII
CLASSIFICATION RESULTS ON THE MUUFL DATASET TO VERIFY THE NECESSITY OF MODALITIES

RS modality couple	OA (%)	AA (%)	κ (÷100)	PR (%)	F1 (÷100)
HSI+HSI	86.76	67.52	82.47	75.58	69.97
LiDAR+LiDAR	53.59	20.11	35.26	25.51	17.07
HSI+LiDAR	**90.03**	**72.61**	**86.80**	**77.45**	**74.09**

TABLE VIII
CLASSIFICATION RESULTS ON THE MUUFL DATASET TO VERIFY THE NECESSITY OF THE ALTERNATIVE OPTIMIZATION PATTERN

Metric	[1, 1]	[1, 10]	[10, 1]	[10, 10]	No Alternation	No Freezing
OA (%)	85.37	**90.03**	83.99	88.97	87.81	89.19
AA (%)	61.98	**72.61**	56.67	70.40	68.66	70.99
κ (÷100)	80.62	**86.80**	78.69	85.35	83.72	85.72
PR (%)	63.43	77.45	72.87	**78.35**	74.98	74.68
F1 (÷100)	61.44	**74.09**	58.43	72.29	70.72	72.31

layers, Conv2 layers, and distribution switches, were removed from the fusion module. Then, the inputs and outputs of the two streams are shown in Fig. 12(c) and (d). Apparently, the corresponding categories in both the inputs are separated. (a3) and (b3) show that HI^{2}D^{2}FNet is capable of narrowing the gap between these categories because the samples from the same category are concentrated relatively. When removing all the alignment-related layers, the clusters from the same category are separated, challenging the classification procedure. For example, the *water* samples are separated into two clusters in (c3). The *building shadow*, *building*, and *sidewalk* samples are scattered in both (c3) and (d3).

D. Ablation Study

Ablation experiments were conducted on the MUUFL dataset to further verify the effectiveness of HI^{2}D^{2}FNet.

1) Necessity of Modalities: The necessity of using dual RS modalities for land-cover classification is comprehensively verified in Table VII. Specifically, *HSI+HSI* and *LiDAR+LiDAR* mean both the modality inputs are HSI and LiDAR patches, respectively. *HSI+LiDAR* is the proposed structure in this article. When only using the HSI, all the classification metrics are slightly decreased since the HSI contains abundant spatial and spectral information. When only using the LiDAR data, these metrics are severely decreased since this modality with one channel only reflects the object elevations. When using both the modalities, higher metrics could be obtained since these modalities are complementary. All the results also verify that the interpretability of HI^{2}D^{2}FNet is specifically presented for HSI and LiDAR data in terms of their RS properties.

2) Necessity of the Alternative Optimization Pattern: The necessity of the alternative optimization pattern is verified in Table VIII. First, we selected the optimal alternative optimization epoch rate within {[1, 1], [1, 10], [10, 1], [10, 10]}. For example, [1, 10] means the decomposition module is optimized for one epoch and then the fusion module is optimized for ten epochs. The results prove that [1, 10] is the best epoch rate for HI^{2}D^{2}FNet, illustrating that the fusion module needs more epochs to be trained. Then, the alternative optimization pattern was removed and all the losses were added together for training. Obviously, all the metrics are decreased since these losses have discrepant optimization directions. Using them together for training a relatively large network with constraints tends to weaken and neglect the contribution of the single loss. Finally, the decomposition module was not frozen when training the fusion module. The obtained results are recorded in the *no freezing* column. Although the results are acceptable to some extent, they are slightly lower than those of HI^{2}D^{2}FNet, which are recorded in the [1, 10] column in Table VIII. Thus, training both the modules separately is beneficial.

3) Necessity of the HI^{2}D^{2}FNet Structure: The necessity of the detailed HI^{2}D^{2}FNet structure is verified in this section. Multiple structure scenarios are as follows.

1) $\mathcal{L}_1$ replaces all the MSA layers with convolution layers.
2) $\mathcal{L}_2$ removes $\log(\cdot)$ and $\exp(\cdot)$, the network structure, and all the losses are accordingly adjusted.
3) $\mathcal{L}_3$ removes the decomposition module and directly uses these two modalities as the inputs of the fusion module.
4) $\mathcal{L}_4$ removes the fusion module and directly concatenates the generated components for classification.
5) $\mathcal{L}_5$ removes the cross-modal geometry stream from the fusion module to prove its effectiveness.
6) $\mathcal{L}_6$ removes the reflectance stream from the fusion module to prove its effectiveness.
7) $\mathcal{L}_7$ removes all the DGC layers.
8) $\mathcal{L}_8$ replaces all the DGC layers with convolution layers.
9) $\mathcal{L}_9$ removes all the distribution switch layers.
10) $\mathcal{L}_{10}$ removes the tradeoff parameter matrices $\boldsymbol{\delta_1}$ and $\boldsymbol{\delta_2}$ and gives both parts the same weight.

The structure of HI^{2}D^{2}FNet is denoted as $\mathcal{P}$. Notably, the aforementioned structures and losses are all adjusted based on their detailed settings. The classification results are listed in Table IX. Specifically, $\mathcal{P}$ outperforms $\mathcal{L}_1$, proving the effectiveness of MSA layers for high-level feature extraction. $\mathcal{L}_2$ generates relatively lower metrics since its structure is not consistent with the classic decomposition steps. These results also verify that the interpretability of HI^{2}D^{2}FNet is enhanced from the intrinsic level. The results of $\mathcal{L}_3$ and $\mathcal{L}_4$ prove that both the modules in HI^{2}D^{2}FNet are significant for accurate classification. Notably, the core mechanism of HIID is directly removed from $\mathcal{L}_3$. $\mathcal{P}$ increases the OA by 2.30% than $\mathcal{L}_3$, illustrating the contribution of the HIID mechanism. The results of $\mathcal{L}_5$ and $\mathcal{L}_6$ show that the proposed guiding strategy is feasible. The cross-modal geometry features are capable of guiding the optimization of the reflectance stream. $\mathcal{L}_7$, $\mathcal{L}_8$, and $\mathcal{L}_9$ reflect that HI^{2}D^{2}FNet aligns cross-modal features from both sample and distribution perspectives. $\mathcal{L}_{10}$ proves the contribution of the proposed tradeoff parameter matrices.

4) Necessity of the Losses: There are several losses in HI^{2}D^{2}FNet. To verify their necessities, they were removed sequentially (i.e., $\mathcal{S}_1 \ldots \mathcal{S}_9$) and the classification results are listed in Table X. It clearly shows that the performance becomes worse if any loss is removed, verifying that the proposed constraints and losses are feasible and suitable for HI^{2}D^{2}FNet. Furthermore, the selection of α is also analyzed in this article. The results with different α values are listed

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TABLE IX
CLASSIFICATION RESULTS ON THE MUUFL DATASET TO VERIFY THE NECESSITY OF THE HI²D²FNET STRUCTURE

Metric	$\mathcal{L}_1$	$\mathcal{L}_2$	$\mathcal{L}_3$	$\mathcal{L}_4$	$\mathcal{L}_5$	$\mathcal{L}_6$	$\mathcal{L}_7$	$\mathcal{L}_8$	$\mathcal{L}_9$	$\mathcal{L}_{10}$	$\mathcal{P}$
OA (%)	87.1	85.3	87.7	84.3	88.2	88.7	88.9	89.15	87.1	88.4	**90.0**
AA (%)	65.2	57.1	65.4	64.5	66.2	71.8	72.7	72.23	65.8	**73.4**	72.6
κ (÷100)	82.8	80.3	83.6	79.2	84.4	85.0	85.3	85.66	83.0	84.8	**86.8**
PR (%)	77.0	66.4	**78.1**	71.5	73.5	74.7	76.5	72.36	70.1	74.9	77.5
F1 (÷100)	68.9	58.1	68.1	66.7	67.2	72.7	73.6	72.21	65.5	73.7	**74.1**

TABLE X
CLASSIFICATION RESULTS ON THE MUUFL DATASET WHEN USING DIFFERENT LOSS COMBINATIONS

Loss	$\mathcal{S}_1$	$\mathcal{S}_2$	$\mathcal{S}_3$	$\mathcal{S}_4$	$\mathcal{S}_5$	$\mathcal{S}_6$	$\mathcal{S}_7$	$\mathcal{S}_8$	$\mathcal{S}_9$	$\mathcal{P}$
L_{REC}	✓	✓	✓	✓	✓	✓	×	×	×	✓
$L_{HSI\to R}$	✓	✓	✓	×	✓	×	✓	×	×	✓
$L_{LiDAR\to S}$	✓	✓	✓	✓	×	×	✓	×	×	✓
L_{DIV}	×	✓	×	✓	✓	✓	✓	✓	×	✓
L_{CON}	✓	×	×	✓	✓	✓	✓	✓	×	✓
L_{CLS}	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
OA (%)	88.3	89.1	88.1	89.5	89.1	87.6	89.0	89.4	88.7	**90.0**
AA (%)	72.0	68.5	70.6	71.2	71.7	65.9	70.1	71.0	72.1	**72.6**
κ (÷100)	84.5	85.6	84.2	86.1	85.5	83.6	85.4	85.9	85.2	**86.8**
PR (%)	73.1	**78.8**	74.5	77.1	76.2	76.0	76.7	76.0	73.0	77.5
F1 (÷100)	71.9	69.4	72.1	72.7	73.2	67.6	71.7	72.1	72.0	**74.1**

TABLE XI
CLASSIFICATION RESULTS ON THE MUUFL DATASET TO ANALYZE THE LOSS WEIGHT α

α	OA (%)	AA (%)	κ (÷100)	PR (%)	F1 (÷100)
0.1	87.89	69.54	83.94	76.21	71.52
1	**90.03**	**72.61**	**86.80**	**77.45**	**74.09**
10	88.66	69.79	84.98	77.04	71.77
100	88.66	69.93	84.99	76.73	71.79

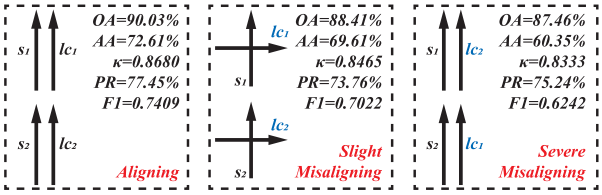

Fig. 13. Performance on the MUUFL dataset in misaligning scenarios.

in Table XI. Notably, α balances the decomposition-related and classification-related losses. The results comprehensively show that the best classification results appear when $\alpha = 1$. Thus, it was selected in this article.

5) Performance in Misaligning Scenarios: In the fusion module, HI²D²FNet is capable of fusing RS modalities from both sample and distribution perspectives by the cross-modal geometry stream. Furthermore, the MSA layers and classifiers contribute to generating acceptable results in misaligning scenarios. Two misaligning scenarios, including slight and severe misaligning scenarios, were specifically designed to verify the performance of HI²D²FNet. In the slight misaligning scenario, the first and second axes of both $\boldsymbol{lc_1}$ and $\boldsymbol{lc_2}$ are directly exchanged. In the severe misaligning scenario, both $\boldsymbol{lc_1}$ and $\boldsymbol{lc_2}$ are directly exchanged. The results are shown in Fig. 13. It proves that the classification performance is still acceptable in these two scenarios, verifying the robustness of HI²D²FNet against the deviation of local information.

TABLE XII
CLASSIFICATION RESULTS ON THE MUUFL DATASET TO SELECT THE OPTIMAL WINDOW SIDE w

w	OA (%)	AA (%)	κ (÷100)	PR (%)	F1 (÷100)	Time/epoch (s)
3	88.14	68.16	84.30	76.42	69.80	**0.62**
5	**90.03**	**72.61**	**86.80**	77.45	**74.09**	0.97
7	88.53	71.21	84.78	**80.35**	73.60	2.01
9	88.29	71.55	84.47	74.80	71.77	3.74
11	85.73	66.41	81.06	69.04	66.83	10.31

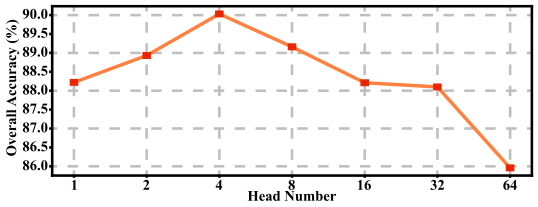

Fig. 14. Classification results on the MUUFL dataset to select the optimal head number.

E. Sliding Window Side Selection

The optimal sliding window side w was selected within $\{3, 5, 7, 9, 11\}$. Small ws cannot provide enough spatial information in the decomposition and fusion procedures. Large ws challenge the analysis of local HSI and LIDAR gradients since more unrelated samples are selected as the neighboring samples of the target sample. Furthermore, large ws make the training process time-consuming. The classification accuracies and training time per epoch are shown in Table XII. Although the shortest training time is needed when $w = 3$, the highest metrics can be obtained when w becomes 5. Then, most metrics tend to be decreased with the increase in w. Thus, $w = 5$ was used in this article.

F. Head Number Selection

The channel number b should be divisible by the head numbers of the MSA layers considering the layer properties. Thus, for the MUUFL dataset, the optimal head number should be selected within $\{1, 2, 4, 8, 16, 32, 64\}$ since $b = 64$. Their classification results in terms of OA are provided in Fig. 14. It clearly shows that the highest OA is obtained when the head number becomes 4. Using more heads makes the whole network hard for training and decreases the OA. Thus, for the MUUFL dataset, all the MSA layers in HI²D²FNet have four heads.

V. CONCLUSION

A multimodal data fusion network called HI²D²FNet has been proposed for hyperspectral and LiDAR classification. Inspired by the core mechanism of HIID, HI²D²FNet aims to extend the application of HIID and preliminarily exploit the potential of this mechanism for multimodal RS data fusion. Furthermore, this technique is capable of exploring the inherent modality connection between HSI and LiDAR data for decreasing the interference of their heterogeneous variance from the intrinsic perspective. Specifically, HI²D²FNet fuses the horizontal geometry information in the shading component

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with the vertical geometry information in the LiDAR data from both sample and distribution perspectives. The obtained cross-modal geometry feature is used to guide the reflectance stream optimization. Multiple cross-modal constraints in terms of prior RS properties have been presented for enhancing the interpretability. The decomposition and fusion modules are optimized in a novel alternative optimization pattern, attempting to achieve a dynamic balance between them. Experiments have been conducted on three datasets to prove the effectiveness of HI^{2}D^{2}FNet. Furthermore, several ablation studies and parameter selection experiments have been provided. The experimental results evaluated by qualitative and quantitative metrics have proved that HI^{2}D^{2}FNet outperforms other techniques. Further experiments will focus on capturing the connection between HSI and LiDAR data from a mathematical perspective.

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